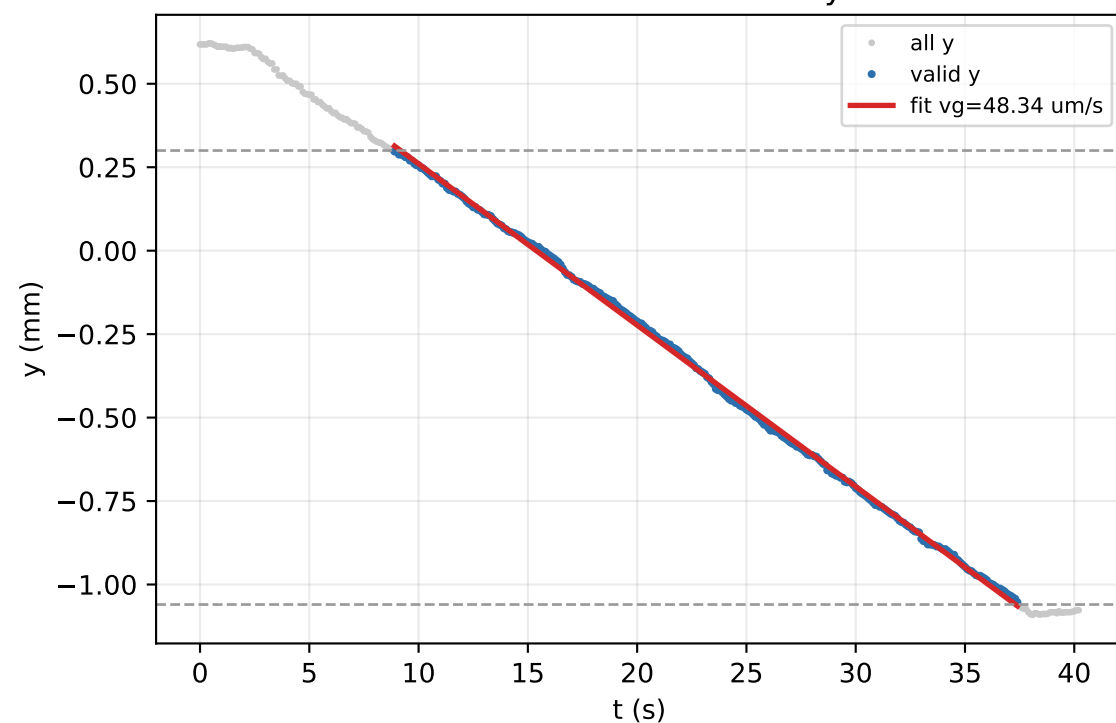
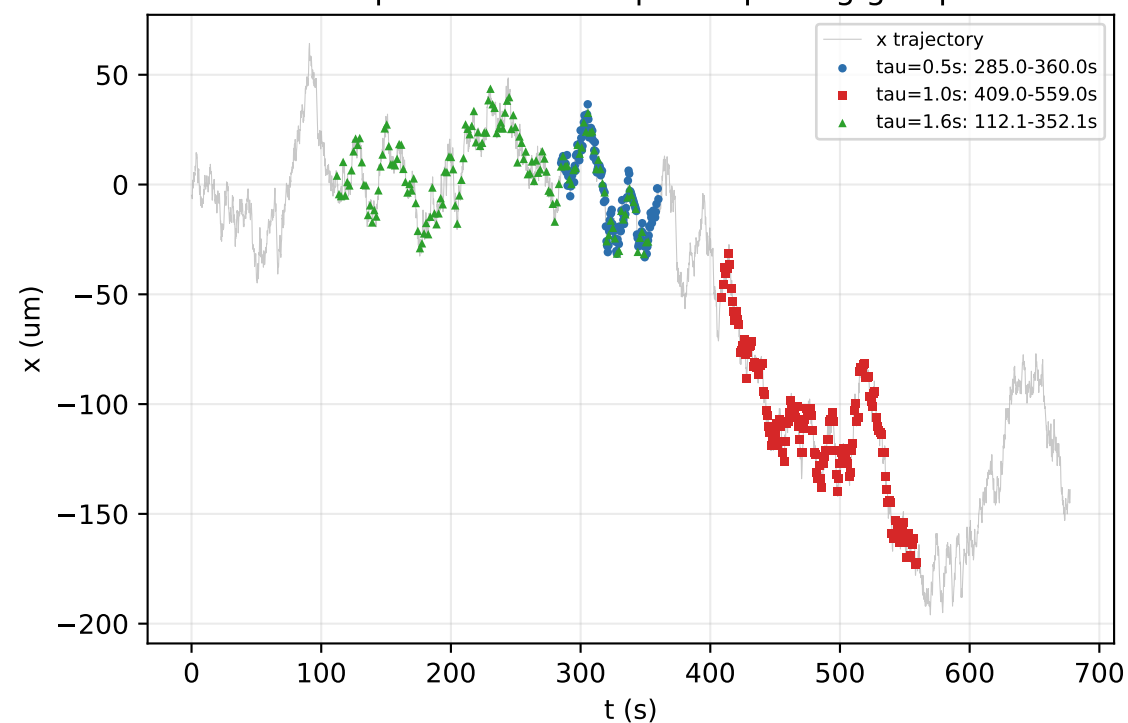


Millikan oil drop particle5: optimized Brownian intervals

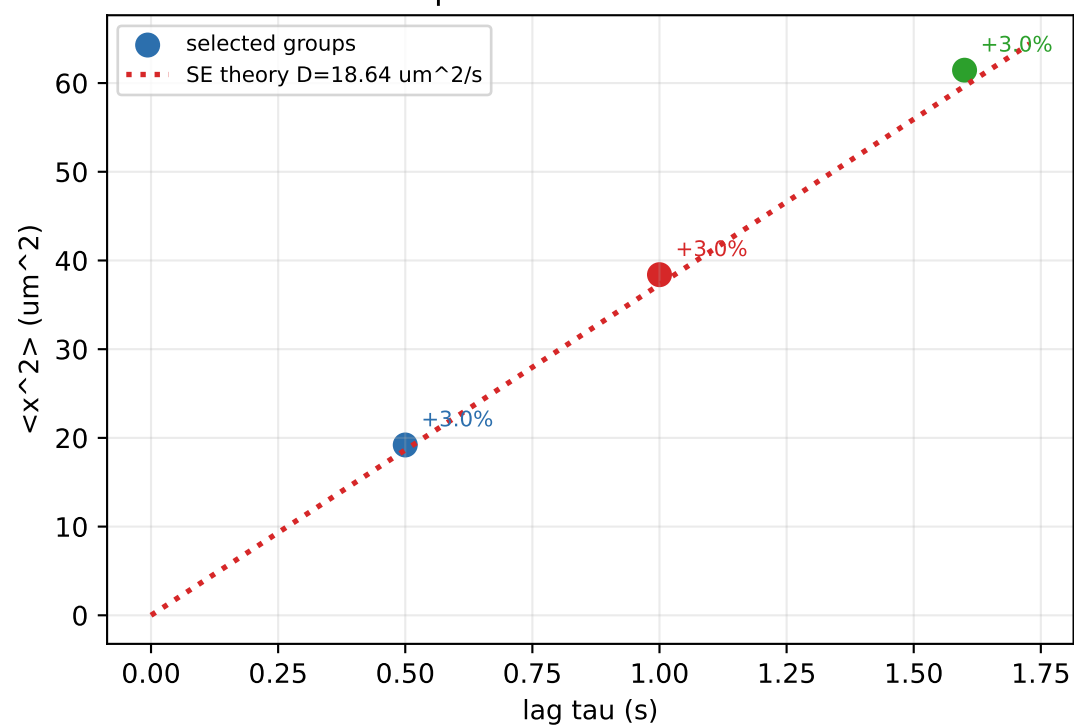
Free-fall terminal velocity



Sampled coordinate points per lag group



Separate $\langle x^2 \rangle$ results



Summary

Particle: particle5
 Window strategy: target_kb
 Freefall valid points: 286
 $vg = 48.3420 \text{ um/s}$, $R^2 = 0.999431$
 $r_{\text{eff}} = 0.6383 \text{ um}$, $d_{\text{eff}} = 1.2767 \text{ um}$
 SE theory $D = 18.6419 \text{ um}^2/\text{s}$

tau	covered time	$\langle x \rangle$	$\langle x^2 \rangle$	k_B err
0.5s	285.0-360.0s	-0.110	19.200	+2.99%
1.0s	409.0-559.0s	-0.804	38.399	+2.99%
1.6s	112.1-352.1s	-0.200	61.450	+3.01%

NIST $k_B = 1.3806e-23 \text{ J/K}$
 No average is reported.